HISABDO AI

DAY 11 — SPENDING PATTERN INTELLIGENCE

Professional MVP Specification & Implementation Guide

1. Document Purpose

This document defines the Day 11 Spending Pattern Intelligence module for HisabDo AI. It analyzes verified transaction data to identify where a user spends money, how spending is distributed across categories, how spending changes between comparable periods, and which recurring or materially changing patterns require attention.

2. Business Objective

HisabDo AI should move beyond recording transactions and provide understandable spending intelligence. The system should answer: Where is the user spending the most? Which category changed the most? How has spending changed compared with the previous period? Which expenses appear recurring?

3. Scope

- Retrieve only the authenticated user's permitted transactions.
- Calculate total expenses and category-wise spending.
- Calculate category percentages and top categories.
- Compare current and previous comparable periods.
- Identify increasing, decreasing, new and stopped categories.
- Detect recurring-spending candidates from transaction history.
- Produce verified structured insights for the AI layer.
- Expose reusable results to mobile and web applications.

4. Required Data

Field	Priority	Purpose
UserId	Required	User isolation
TransactionId	Required	Unique transaction
TransactionType	Required	Income/Expense separation
Amount	Required	Financial calculations
Category	Required	Category analysis
Date	Required	Period comparison
Description/Note	Recommended	Recurring-pattern detection
Customer/Person	Optional	Future contextual analysis

5. Analysis Period

The current period and previous period must be comparable. For a custom range, the previous period should contain the same number of days. For complete calendar months, compare one month with the immediately preceding month. The API must return both periods.

6. Spending Calculations

Total Expense = Sum of valid expense transaction amounts within the selected period.

Category Amount = Sum of expense amounts belonging to that category.

Category Percentage = (Category Amount / Total Expense) × 100.

Categories are sorted by amount descending. If total expense is zero, percentages are zero and no division-by-zero error is allowed.

7. Period Comparison

Growth % = ((Current Expense − Previous Expense) / Previous Expense) × 100.

- Negative growth means spending decreased.
- Zero means spending is unchanged.
- Positive growth means spending increased.
- If previous expense is zero, return null rather than dividing by zero.
- Apply the same comparison to categories where possible.

8. Category Trend Classification

- Increased — current amount is higher.
- Decreased — current amount is lower.
- Unchanged — amounts are equal.
- New — current spending exists but previous spending was zero.
- Stopped — previous spending existed but current spending is zero.

9. Recurring Spending Detection — MVP

Recurring spending is a candidate pattern, not a guaranteed subscription classification.

- Group repeated descriptions/merchants when available.
- Require multiple historical occurrences.
- Compare transaction dates for approximately regular intervals.
- Use average amounts rather than requiring exact equality.
- Return a candidate/qualification result instead of claiming certainty.
- Production iterations may add merchant normalization, interval tolerance, amount tolerance and confidence scoring.

10. Verified Insight Payload

- Total expense
- Top category, amount and percentage
- Previous-period total
- Expense growth percentage
- Largest increasing category
- Largest decreasing category
- New/stopped categories

- Recurring-spending candidates

AI may explain these verified facts, but it must not invent amounts, percentages, categories or trends.

Flow: HisabDo Database → Analytics Engine → Verified Spending Patterns → LLM Explanation → User

11. Proposed API

GET /api/ai/spending-patterns/{userId}?from=2026-09-01&to=2026-09-30

Example response:


```
{
  "userId": "user-123",
  "period": {"from": "2026-09-01", "to": "2026-09-30"},
  "previousPeriod": {"from": "2026-08-02", "to": "2026-08-31"},
  "totalExpense": 108500.00,
  "previousTotalExpense": 96000.00,
  "expenseGrowthPercent": 13.02,
  "topCategories": [
    {"category":"Food","amount":32000.00,"percentage":29.49},
    {"category":"Transport","amount":18500.00,"percentage":17.05},
    {"category":"Shopping","amount":15000.00,"percentage":13.82}
  ],
  "largestIncreasingCategory": {
    "category":"Food","changeAmount":5000.00,"changePercent":18.52
  },
  "recurringCandidates": [
    {"description":"Internet","averageAmount":2500.00,"frequency":"Monthly"}
  ]
}
```


12. Backend Processing Flow

- Authenticate request.
- Validate date range.
- Retrieve only the user's permitted transactions.
- Filter valid expenses.
- Calculate total expense.
- Calculate category amounts and percentages.
- Calculate previous comparable period.
- Calculate total and category changes.
- Detect recurring-spending candidates.
- Build verified insights.
- Return API response.

13. Mobile & Web Output

- Total Spending
- Top Spending Categories

- Category Percentages
- Current vs Previous Period
- Expense Growth
- Largest Increasing/Decreasing Category
- Recurring Spending
- AI Spending Insight

14. Edge Cases & Testing

- No transactions.
- No expense transactions.
- Total expense = 0.
- Previous period expense = 0.
- Missing/invalid category.
- Duplicate transaction.
- Negative/invalid amount.
- Very large transaction.
- Only one period of history.
- Different period lengths.
- Unauthorized access to another user's data.
- Verify AI uses only supplied verified values.

15. Team Deliverables

- Approved requirements.
- Backend spending analytics service.
- Spending Pattern API.
- Category and trend calculation logic.
- Recurring-spending detection logic.
- Verified insight payload.
- AI explanation handler.
- Mobile/Web integration.
- Test cases and results.
- Updated documentation.

16. Team Lead Acceptance Checklist

- Calculations match source transactions.
- Comparable periods are correctly defined.
- Category percentages are correct.
- Trend classification is correct.
- Recurring results are treated as candidates.
- AI receives verified values only.
- Authorization/data isolation is enforced.
- Edge cases are handled.
- API works for mobile and web.
- Day 11 is validated against sample data.

17. Implementation Note

This specification does not assume an undocumented HisabDo database schema. Map the DTOs, repository, authentication and API conventions in the supplied codebase before production merge.

18. Final Definition

HisabDo Spending Pattern Intelligence is an explainable analytics module that uses verified transaction data to identify total spending, category distribution, period-over-period changes and recurring-spending candidates. The analytics engine determines financial facts; AI explains those verified facts in natural language.